

FERINTOSH AGCM, AB, Canada

WMO: 716730

Lat: 52.7439N

Lon: 112.8580W

Elev: 2608

StdP: 13.36

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-22.6	-17.5	-28.6	1.2	-22.3	-23.3	1.7	-17.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.1	81.7	62.1	78.9	61.4	76.1	60.0	66.9	76.5	64.6	74.5	62.6	72.1	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.3	95.8	72.7	60.7	87.4	69.6	58.7	81.1	67.5	33.0	76.1	31.1	74.6	29.5	72.1	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-29.4	87.5	5.6	3.9	-33.4	90.3	-36.7	92.5	-39.9	94.7	-44.0	97.5		
(5)			-29.6	71.2	5.6	2.8	-33.6	73.3	-36.9	74.9	-40.0	76.5	-44.0	78.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.9	15.1	11.8	23.2	37.0	50.8	57.5	60.7	59.3	51.1	38.6	23.6	12.8		
	DBStd	20.91	13.89	14.61	13.98	9.67	8.12	5.26	4.76	5.46	7.98	8.63	12.81	13.32		
	HDD50	5877	1083	1070	832	399	91	8	0	3	81	364	794	1152		
	HDD65	10280	1548	1490	1297	840	443	229	147	188	420	818	1243	1617		
	CDD50	1107	0	0	0	9	115	234	333	291	113	11	1	0		
	CDD65	36	0	0	0	0	2	5	15	11	3	0	0	0		
	CDH74	1048	0	0	0	4	120	131	293	319	176	5	0	0		
	CDH80	155	0	0	0	0	15	8	38	54	40	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	16.7	0.8	0.5	0.7	1.2	1.9	3.1	3.0	2.2	1.3	0.7	0.6	0.6		
(15) Precipitation	PrecMax	20.8	2.5	1.0	1.5	2.4	3.3	4.5	6.5	4.2	3.0	1.8	2.2	1.2		
	PrecMin	11.4	0.0	0.1	0.1	0.2	0.6	1.0	0.9	0.2	0.1	0.2	0.1	0.2		
	PrecStd	2.8	0.6	0.3	0.4	0.6	0.8	1.0	1.3	0.9	0.8	0.4	0.5	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.9	43.2	55.6	72.3	82.3	80.7	84.2	83.9	85.2	72.4	58.5	43.0		
		MCWB	36.9	36.6	45.0	50.2	54.9	61.1	68.3	62.3	59.5	53.7	45.2	37.1		
	2%	DB	41.0	38.8	48.9	65.1	77.5	78.0	80.4	81.2	79.2	63.6	49.2	39.0		
		MCWB	35.8	33.8	40.2	47.4	53.4	59.7	65.6	63.6	57.6	49.5	40.5	34.0		
	5%	DB	38.1	35.7	43.6	58.8	73.0	74.6	77.5	77.7	74.1	58.4	43.5	34.9		
		MCWB	33.9	32.0	37.4	44.0	52.3	58.1	64.0	63.1	56.1	46.1	37.0	30.7		
	10%	DB	34.7	32.5	39.5	53.5	68.2	71.0	74.2	74.1	68.2	53.3	39.5	30.5		
		MCWB	31.2	29.7	34.5	41.7	50.2	56.3	62.4	61.7	53.9	43.7	34.7	27.4		
	0.4%	WB	37.8	37.0	45.2	51.2	57.0	65.6	71.4	69.5	61.8	54.6	45.8	37.9		
		MCDB	43.1	42.4	53.7	70.8	74.4	76.2	80.7	78.1	78.1	70.3	57.7	41.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.9	34.5	40.6	47.7	55.0	62.6	67.7	66.6	59.6	50.6	41.2	34.1		
		MCDB	40.4	38.2	47.8	63.9	72.5	73.1	76.9	76.5	75.3	60.4	47.9	38.5		
	5%	WB	34.0	32.1	37.5	44.8	53.4	60.3	65.3	64.4	57.0	47.5	37.8	31.1		
		MCDB	37.8	35.1	43.2	56.5	69.5	69.6	74.8	74.7	70.8	56.5	42.5	34.7		
	10%	WB	31.3	29.5	34.8	42.2	51.6	58.1	63.3	62.2	54.9	44.7	35.2	27.6		
		MCDB	34.6	32.6	39.0	52.4	65.9	67.6	72.4	71.8	67.6	52.6	38.7	30.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.2	19.2	18.9	21.5	24.6	22.1	23.1	25.0	25.6	21.7	17.9	16.5		
		MCDBR	19.3	20.9	23.0	32.2	32.1	28.3	28.9	32.0	37.1	30.6	22.5	18.9		
		MCWBR	15.3	16.9	15.4	16.6	13.7	13.1	16.6	17.2	17.8	17.1	15.0	14.8		
		MCDBR	19.8	20.0	22.2	29.5	28.0	24.3	26.1	28.0	32.7	28.4	20.8	18.7		
	5% WB	MCWBR	16.1	16.4	15.4	15.9	12.6	13.1	15.9	16.5	16.7	16.4	14.6	15.0		
(40) Clear-Sky Solar Irradiance	taub		0.263	0.264	0.294	0.325	0.343	0.352	0.351	0.362	0.310	0.281	0.253	0.246		
	taud		2.290	2.386	2.401	2.377	2.382	2.377	2.413	2.354	2.478	2.535	2.461	2.306		
	Ebn at Noon		233	275	287	289	288	285	283	273	278	267	243	220		
	Edh at Noon		19	24	29	34	36	36	34	34	27	20	16	16		
(44) All-Sky Solar Radiation	RadAvg		335	668	1125	1494	1824	1925	2004	1640	1136	672	337	245		
	RadStd		33	55	83	142	119	124	103	113	121	79	36	23		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (88 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page